

# Scale-Invariant Field Organization as a Subset of the Universal Quantum Foam Hypothesis: Atoms and Galaxies as Homologous Field-Mechanical Realizations

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## Abstract

This preprint addresses a specific sub-area of the Universal Quantum Foam Hypothesis (UQSH) and focuses on the mechanical symmetry between micro- and macrocosmic structures. While the basic conceptual and ontological framework of the UQSH was already presented in the abstract of January 6, 2026 (based on the paper from November 2025), this work focuses on overcoming the fundamental separation

between the quantum world and the cosmos. At its core is the thesis that atoms and galaxies are not isolated systems, but rather functionally homologous realizations of the same field-mechanical processes at different scales. In this extension, the atom is defined not as a collection of solid particles, but as a local field node, a zone of increased field tension in which motion is structured and held. Electrons are interpreted as stable vortex modes of the medium, whose discrete states result from the mechanical stability conditions of the field. This multi-scale perspective establishes the field's reorganization rate as the basis for understanding time and entropy. Entropy is understood here as a process of rearranging organization between bound patterns and free field dynamics. This subfield of UQSH complements the existing framework with a perspective that closes the interpretive gap between the Planck scale and the cosmic macrostructure through the principle of scale-invariant field mechanics.

## **1 Scale-Invariant Field Organization – Atoms and Galaxies as Expressions of the Same Principle**

The Universal Quantum Foam Hypothesis (UQSH) posits that nature recognizes no fundamental separation between "microcosm" and "macrocosm." What we call the quantum world and what we call the cosmic world are not separate levels of reality, but rather different scales of the same field-mechanical process. Atoms and galaxies are not merely metaphorically comparable; they are functionally homologous realizations of the same field organization.

In this view, an atom is not a collection of solid particles, but a local field node: a zone of increased field potential where motion does not proceed freely, but is contained and structured. The "nucleus" is not primarily an object, but a reference anchor, a maximum density of the field. Electrons are not small spheres orbiting this nucleus, but stable vortex modes of the field that can only exist under specific voltage conditions. They are bound forms of motion, not things.

That electrons do not exist continuously, but only in discrete states, is not a quantum mechanical accident, but an expression of field mechanics: Only certain vortex geometries are sufficiently low in voltage to be permanently supported by the medium. Intermediate states would be unstable. The electron does not "jump" because it is mysterious, but because the field only allows patterns that are fully locked in place. This structure repeats itself on a cosmic scale. Supermassive black holes are not exotic anomalies, but the macroscopic equivalent of atomic reference nuclei: maximum field knots where motion no longer propagates freely. Galaxies form extended potential spaces around these knots, comparable to atomic shells. Within these spaces, bound vortex states arise: stars, gas clouds, and especially dwarf galaxies.

Dwarf galaxies exhibit behavior strikingly reminiscent of electron states. They appear in galactic halos, are incorporated, merge, are expelled, or lose their identity. In many cases, they appear suddenly, as if arising out of nowhere, often triggered by interactions, collisions, or strong field gradients in the vicinity of large galaxies or black holes. Likewise, they can disappear, dissolve, or be integrated. In the UQSH, this is not random but the macroscopic equivalent of what is observed on an atomic scale as the creation and disappearance of electron states. Electrons do not appear because "particles are created" but because the field, under certain conditions, allows a new stable vortex geometry. Similarly, dwarf galaxies do not form because matter clumps together randomly, but because the

galactic field develops a new stable vortex state.

Matter formation is thus the same process at all scales: free field motion is transformed into bound patterns under increasing tension. At the quantum level, this occurs at high energies, strong gradients, and narrow space-time scales. At the cosmic level, it occurs at massive field nodes, accretion events, and large-scale tension axes. The mechanism is identical; only the scale differs. This view also explains why black holes play such a central role in cosmic evolution. They are not endpoints but generators of structure. Jets, shock waves, density gradients, and magnetic orders arise in their vicinity, enabling new vortex states. Galaxies do not form despite black holes but because of them, just as atomic structures exist not despite but because of nuclear compression.

The analogy extends further: In quantum mechanics, electrons often appear in pairs as  $e$  and  $e$ . This pairing is not an arbitrary symmetry, but rather the field-mechanical consequence of any vortex formation: A stable vortex necessarily generates a counter-vortex. The field cannot support unidirectional rotation. On a cosmic scale, a similar principle is evident in the observed rotation of galaxies: The majority rotate preferentially in one direction, a significant proportion in the opposite direction. This asymmetry is not a cosmological accident, but rather an expression of large-scale vortex formation in the field, in which counter-structures necessarily arise.

A single atom is therefore not a "miniature version" of a galaxy; rather, both are manifestations of the same law: stability arises not from equilibrium, but from controlled inequality. A field node establishes a reference state. Around it, vortices form that must neither collapse nor escape. They must remain within a narrow voltage range, precisely where stable patterns are possible. From this perspective, an electron is functionally closer to a dwarf galaxy than to a point object. Both are bound field modes within a larger potential space. Both exist only as long as

the field conditions support their geometry. Both can emerge, disappear, and rearrange themselves. And both are not isolated, but always part of a larger field system.

The UQSH thus makes a radical statement: There are no elementary building blocks of reality. There are only stable patterns of the field. Atoms are not fundamental units; they are local solutions. Galaxies are not special cases; they are the same solutions on a larger scale. Matter is not substance, but congealed motion. Structure is not random, but rather what a field-based universe inevitably produces. Thus, the universe no longer appears as a collection of different levels, quanta here, stars there, but as a continuous process. What appears in the laboratory as a quantum leap is visible cosmically as galaxy formation. What appears in the telescope as galactic reorganization is realized atomically as a rearrangement of states. The separation between microphysics and astrophysics is therefore not ontological, but only epistemic. It arises from our scale, not from the structure of the world.

## **2 Counter-vortex and stability – Why order always needs two directions**

This perspective explains why counter-vortices do not appear as errors or exceptions, but rather as a necessary component of any stable order. A field that allowed only one direction of rotation would build up momentum, not provide support. Every stable vortex formation therefore inevitably generates a complementary structure. At the quantum level, this appears as the pairing of  $e$  and  $e$ . At the cosmic level, it manifests in the statistical distribution of galactic rotations, in counter-rotating filament flows, and in large-scale shear surfaces of the cosmic web.

Crucially, the counter-structure is not an antiverse, "not an alien state,

but the same field process in the opposite phase. Both vortices belong to one and the same medium. They are not enemies, but partners in a dynamic equilibrium. Only their coexistence allows the field to bear tension without collapsing.

The observed asymmetry for example, that two-thirds of galaxies preferentially rotate in one direction and one-third in the opposite direction, is not, from this perspective, a coincidence but rather an expression of a global field organization. A perfectly symmetrical relationship would be unstable, a purely one-sided one impossible. The field requires a preponderance to define direction and, at the same time, a counterweight to prevent this direction from tipping into resonance. Stability arises not from equality but from controlled inequality. This also explains why triplet constellations occur on all scales. Two vortices generate tension. Only a third state allows this tension to be spatially distributed. At the atomic level, this manifests as the necessity of additional degrees of freedom: position, momentum, and phase. At the cosmic level, it appears as groups of three galaxies, as clusters consisting of a main galaxy, companions, and transition regions, as filamentary intersections with three dominant directions. Three is not symbolic but mechanical: it is the smallest number with which a field can simultaneously possess orientation, depth, and stability.

In this sense, the existence of long-lived positrons is not a contradiction. A counter-vortex is not inherently unstable. It only becomes annihilating when it is coherently superimposed with its complementary partner. As long as both are embedded in separate field spaces, they can exist, move, and interact. Only when their phases coincide, microscopically through direct encounter, macroscopically through large-scale superposition, does the field lose the need to keep them separate. The order dissolves. Macroscopically, this means that even galaxies with opposite rotation are not "cosmic positrons" destined to disappear. They are necessary counter-

structures within a global vortex system. Only when two such systems come into coherent superposition, for example in a dense, counter-rotating merger, does an annihilation-like process begin: loss of coherent rotation, massive energy release, transition to diffuse field motion, and the formation of new patterns.

Annihilation is therefore scale-invariant. It is not an event, but a transitional state:

- At the quantum level: release into gamma rays.
- At the stellar level: supernova, accretion collapse, radiation burst
- At the galactic level: merger, field decoupling, cosmic reorganization.
- At the cosmic level: large-scale redistribution of order into free motion.

In all cases, the same rule applies:

Structure exists only as long as the field is forced to maintain local tension.

As soon as this tension cancels itself out, order becomes superfluous.

The universe doesn't "lose anything" in the process. It merely changes the state of its motion. Matter is congealed dynamics. Radiation is released dynamics. Annihilation is the transition between the two. This explains why counter-vortices appear not as an anomaly, but as a necessary condition of existence. Without them, there would be no duration, no depth, no hierarchy. A universe without counter-structure would be either rigid or chaotic, but never viable. The UQSH therefore describes the universe not as an arena of particles, but as a continuous dialogue of the field with itself: order and counter-order, binding and release, pattern and unfolding. Every scale repeats the same grammar. Only the pace changes.

It is crucial not to confuse these levels. When the UQSH refers to the

electron and its countervortex, the positron, it is not referring to two ontologically separate particle types, but rather to two complementary orientations of the same field-mechanical pattern. Large-scale sky surveys show that the rotations of galaxies are not entirely randomly distributed. Across cosmic scales, weak but systematic asymmetries emerge: In certain regions, preferred rotation directions dominate, while neighboring structures develop opposing patterns. Directed flows of matter and large-scale shear surfaces, in which vortex and counter-vortex structures arise, can also be observed within the cosmic web itself. Galaxies do not form in isolation but are embedded within these flows. Their rotation is therefore part of a coupled system. According to the UQSH (University of the Quantum of Solar System Research), these findings are not coincidental but macroscopic signatures of a field-wide vortex dynamic: Every dominant motion necessarily generates a counter-structure. Asymmetries, counter-rotations, and shear surfaces are expressions of a global, multi-scale field organization.

If this principle is consistently applied to the atomic scale, an expanded picture of the structure of matter emerges. A stable atomic pattern would therefore not be a binary system of nucleus and electron, but rather a minimal vortex network of three coupled field states. Two of these states carry the dominant strain state; they define direction, bonding, and reference. The third forms the complementary counter-structure, which prevents this order from tipping into resonance or collapsing. In the classical description, these states appear to us as individual electrons. However, from a field-mechanical perspective, they would not be independent particles, but rather projections of a three-membered vortex node. The countervortex need not be permanently present as a freely measurable positron. It can be locally bound, phase-shifted, or energetically suppressed, structurally present without becoming visible as a separate particle. Only under strong excitation, collision, or high field gradients can this counter-component decouple and appear as a real positron. An-

nihilation in this sense is not the disappearance of two foreign objects, but the return of a split vortex pair to free field motion. Electrons would thus not be miscounted, but misunderstood. What appears as a single particle is the visible component of an asymmetric triadic system whose internal counterstructure usually remains hidden. Thus, the same rule repeats itself at all scales: stability arises not from pairs, but from triadic relationships in which predominance and counterstructure coexist.

### 3 Entropy across scales

In UQSH, entropy can be formulated, in terms of scale, as a measure of the inaccessibility of field-internal degrees of freedom relative to an observer's scale. A "closed" entropy budget does not arise from the fact that the universe is finite, but rather from the fact that each scale can only count a limited portion of the field-mechanical dynamics as a "state." What lies below a scale appears as effective noise or as unresolvable microdynamics; what lies above appears as a quasi-constant boundary condition or as a slowly varying background. In this sense, entropy is not an absolute quantity of the field, but a scaled balance: it counts how many field-mechanical patterns are distinguishable and reconstructible for a given scale.

Starting from the lower boundary scale below the observer (formally: "lower scale," extending to Planck-like dynamics), a source of entropy arises through continuous reorganization: The field produces high-frequency, short-lived degrees of freedom that the observer cannot hold as ordered information and therefore appear as entropic "loss channels." Starting from the upper boundary scale above the observer (galaxies, filaments, large-scale vortex organization), a second, complementary boundary emerges, mirrored: The large-scale field order acts as a slow boundary condition that "closes" state spaces by forcing local patterns into long-term flow lines.

Thus, in the UQSH, entropy becomes a bridging variable between scales: It is the balance of information that is not carried along during the transition from free field motion to bound pattern form (and back) across scales.

If this logic is extended beyond the connecting scales, a scale-hierarchical, in principle infinite field emerges, in which "closedness" is understood not as spatial finitude, but as the self-consistency of scale coupling. In an infinite universe, entropy can nevertheless be globally "closed" if each scale possesses its seemingly entropic outflows as inflows into neighboring scales: order on one scale is then always bought at the price of disorder on another, and vice versa. The field as a whole remains reversible in the sense of its overall organization, while observers on finite scales measure irreversible entropy increases because their resolution fundamentally precludes the complete reconstruction of the scale flows. Thus, in the UQSH, entropy is the scale-dependent shadow of an infinite, but closed, field economy: locally irreversible, globally self-consistent.

## 4 Cross-scale entropy and closed dynamics

Within the framework of UQSH, entropy does not appear as a measure of increasing chaos, but rather as a field-mechanical tendency toward stress relaxation. Every stable pattern, from the electron to the galaxy, represents a local increase in field stress. Entropy is the universal motion of this medium to balance these stresses without fundamentally destroying organization. This results in a closed dynamic across all scales: at the lower scale, high-stress patterns are converted back into free field excitation through complementary coupling; at the upper scale, saturated large nodes, such as black holes, become storage and redirection centers for the same field work. The cosmic microwave background radiation does not mark a final state, but rather a long-lived, circulating state of free excitation that can transition back into denser regimes at any time

along stress gradients. Entropy thus does not signify "loss," but rather a redistribution of organization between bound and free states.

In this view, time is not an external parameter, but rather the measure of the field's reorganization rate. Where field tension is high, this rate slows down; where it is low, it accelerates. The entropy arrow follows inevitably from this: reorganization always occurs in the direction of lower tension difference, not because order is impossible, but because every new order generates tension anew. The universe thus forms a closed system across scales, in which neither energy nor organization disappears, but rather alternates cyclically between pattern binding and free dynamics. The lower scale is not a "beginning," the upper not an "end"; both are limiting regimes of the same medium. Entropy does not describe the disintegration of the world, but rather the rhythm with which a single field distributes its tensions, from the Planck scale to the cosmic macrostructure and beyond.

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## Zitat

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